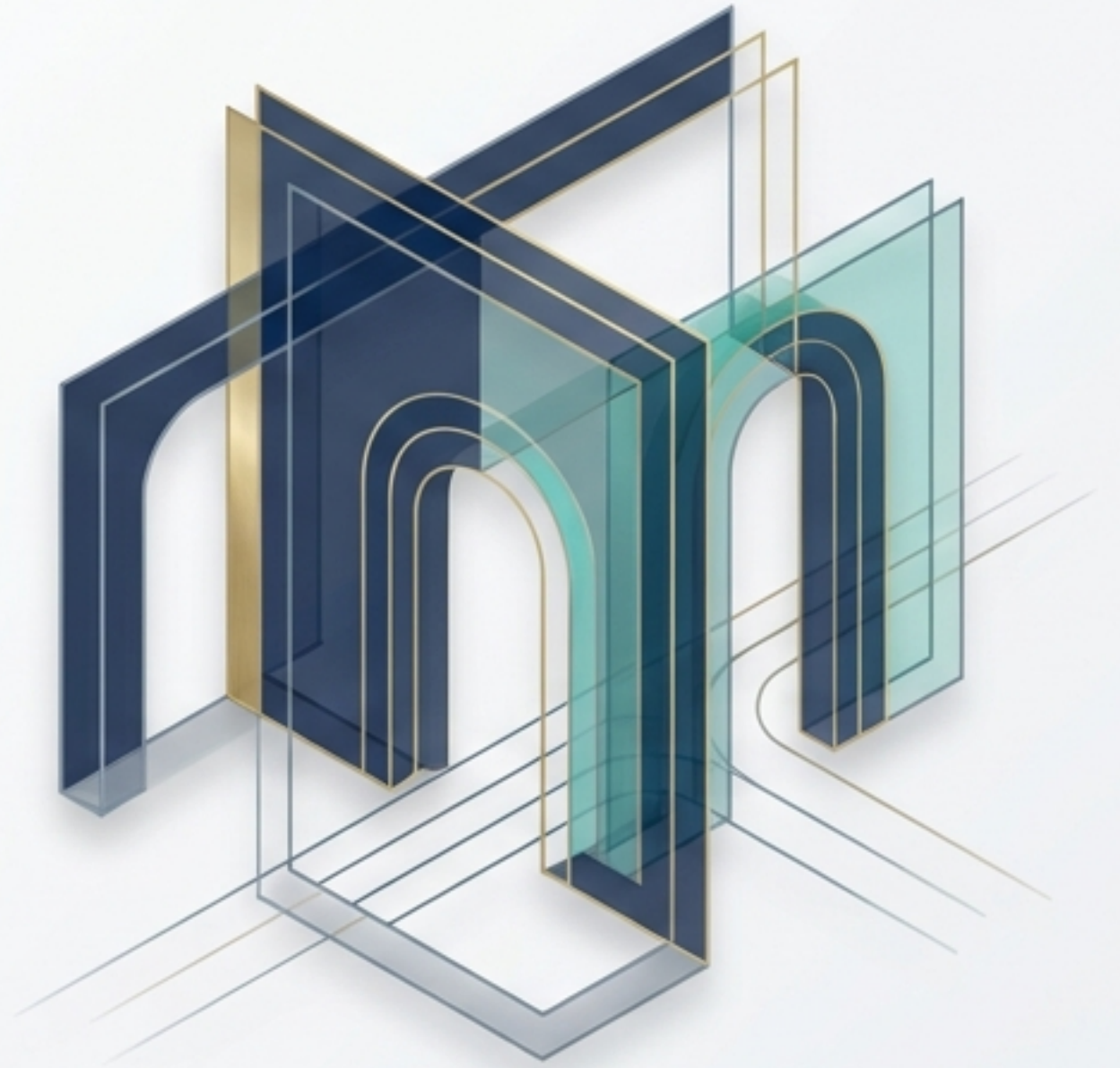


The Enterprise SDLC Compliance Agent

Governing automated SDLC quality gates.



The Review Bottleneck and the Case for Deterministic Evaluation

The Bottleneck & The Bar

70–100 Pages

High turnaround manual reviews
by a small pool of EA architects.

The System Mandate

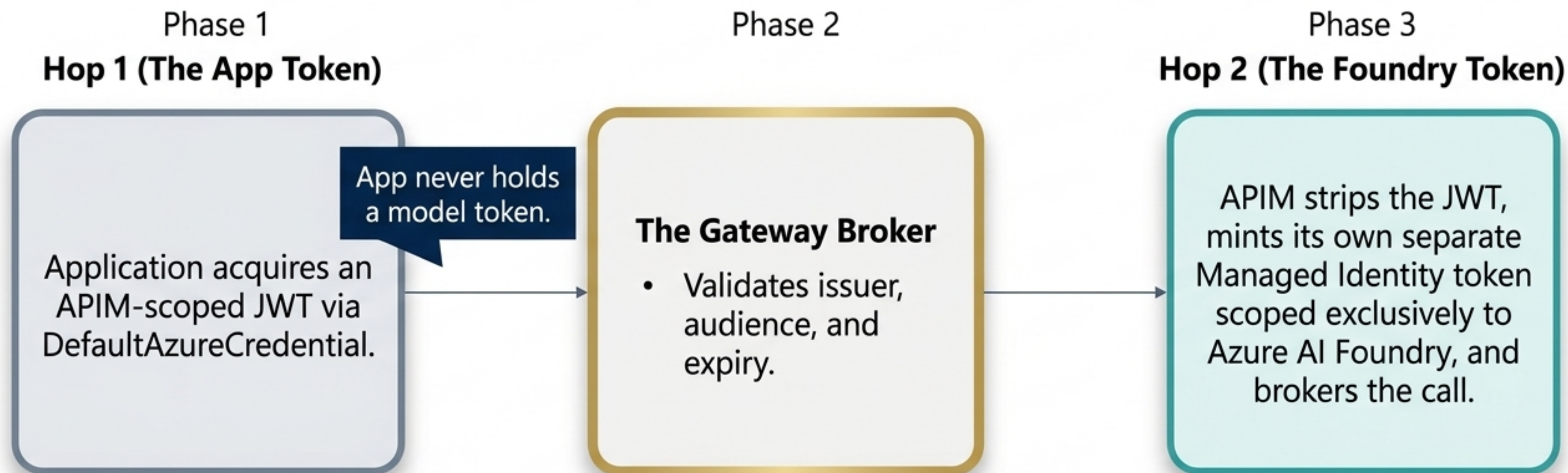
- ✓ Trace gaps to real clauses.
- ✓ Cover the entire document.
- ✓ Zero hallucinations.

Diagnostic Matrix

Why Deterministic Rubrics Beat RAG

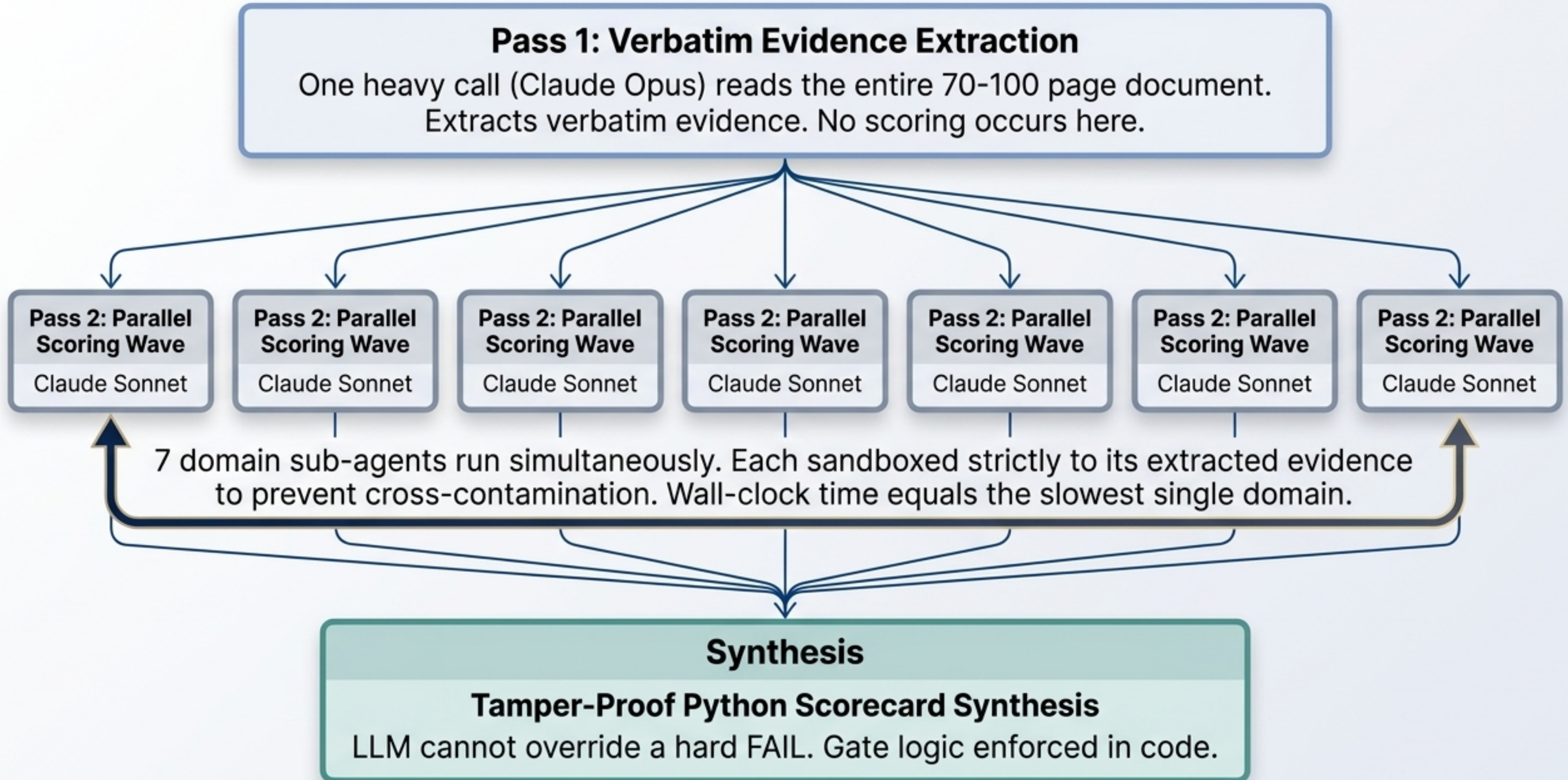
	Rubrics	RAG
Scope	✓ Evaluates the single submitted document whole	✗ Built on a corpus via search
Retrieval	✓ Deterministic, no top-k subset	✗ Top-p/Top-k where chunking drops content
Semantic	✓ Rubric pre-defines good/bad	✗ Unsupervised relative understanding
Traceability	✓ Exact clause citation	✗ Depends on chunk boundaries
Cost	✓ Token cost only	✗ Tokens + Search + Embeddings compute
Security	✓ Uses existing model deployment	✗ Requires standing up new Azure resources

Zero Standing Secrets via Dual-Token Gateway Brokering



Tokens never merge. The caller's badge opens the gateway; the gateway's badge opens the model.

Two-Pass Orchestration Eliminates Evidence Drift



Slashing Protocol Waste via Asymmetric Turn Optimizations

Optimization 1: Capping Turn-2 Output

31,000 → 13 Tokens

Turn 2 is a protocol handshake, not a reasoning step. Capping output at max_tokens=1 collapsed wasted summary tokens, saving \$0.47 per run.

Optimization 2: Prefix Warming

7 Writes → 1 Write

Firing one minimal call before the parallel wave ensures all 7 sub-agents arrive to a warm cache. Saved \$0.19 per run.

Optimization 3: Cache Breakpoint Shift

71.8% Peak Hit Rate

Shifted the Pass-1 cache breakpoint onto the document body, bypassing the 1,024-token floor and unlocking massive cross-turn caching.

Independent Convergence on Industry Best Practices

Uber Engineering Parallels



Turn-based prompt caching to avoid re-paying repeated history.



Sub-agents defaulting to a cheaper model under a stronger primary model.



Org-wide spend tracking and session-level telemetry dashboards.

The Enterprise SDLC Compliance Agent



System prompt + document caching; Turn 2 capped at max_tokens=1.



Pass 1 heavier extraction (Opus), delegating to cheaper parallel sub-agents (Sonnet).



Call-level telemetry and exact unit economics rendered in standalone HTML reports.

Zero-API Pre-Flight Cost Estimation



Input

Reads offline file metadata (page counts, image pixel dimensions) in milliseconds.



Processing Engine

Zero API Calls.

Derives token scale and routes model selection automatically based on offline Python calculations.



High-Fidelity Output

Generates Expected Low/Mid/High cost brackets.

Calibrated
within 2%
(\$0.0002)
of actual Anthropic
invoices.

Separation of Concerns in Enterprise AI Governance

Dual-Pillar Ledger

Enterprise Architecture Ownership



rubric.yaml

Controls the business logic (domains, criteria, thresholds). Completely separate from operations.

Engineering Ownership



scorer_config.yaml

Controls operations (model aliases, runtime tuning).

Shared Governance Layers

Immutable Prompts

Versioned like software (MAJOR/MINOR) with full audit-trail stamping and one-line rollbacks.

Inline Guardrails

APIM Content Safety (llm-content-safety) applied inline inbound/outbound, closing native platform gaps.

Business & Technical Impact at Portfolio Scale

Quality

2 → 9

Portfolio shift of passing documents,
reaching decision-grade quality.

Efficiency

68% ↓

Wall-clock latency reduction
(363s to 117s).

Cost Reduction

45% ↓

Total cost reduction across
tracked documents.

Unit Economics

\$0.66

Mean cost per scorecard
across 33 production runs.

THE ENTERPRISE SDLC COMPLIANCE AGENT — AUDITABLE, DETERMINISTIC, AND HIGHLY GOVERNED.